Error Sources: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

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Solution 1 — Random versus systematic. Return →

Photon noise and scintillation are mainly random. A ruler zero error and a wrong filter zero point are systematic. **Check:** random errors scatter around a mean; systematic errors shift the mean.

Solution 2 — Averaging. Return →

For 9 readings, the standard error is $0.3/3 = 0.1$. To reach 0.05, $n = (0.3/0.05)^2 = 36$. **Check:** halving the uncertainty requires four times the readings.

Solution 3 — Bias does not average. Return →

The 0.2 kg bias remains in the mean. Averaging reduces random scatter but not a fixed offset. The remedy is to calibrate the scale and subtract the offset, or include it as a systematic uncertainty. **Check:** 100 wrong zeros still give a wrong zero.

Solution 4 — Combined uncertainty. Return →

The combined uncertainty is $\sqrt{0.03^2 + 0.04^2} = 0.05$ mag. **Check:** independent uncertainties add in quadrature; the larger term dominates.

Solution 5 — Photon noise. Return →

The uncertainty is $\sqrt{10000} = 100$ photons. The fractional uncertainty is $100/10000 = 0.01$, or 1%. **Check:** photon noise decreases as $1/\sqrt{N}$.

Solution 6 — Precision and accuracy. Return →

The measurements are precise but not accurate. A systematic error is likely, such as a calibration offset or wrong model. **Check:** tight clustering means small random scatter, not correctness.

Solution 7 — Extinction error. Return →

The uncertainty is $0.02 \times 2 = 0.04$ mag. **Check:** extinction errors grow with airmass, so low-altitude observations are more vulnerable.

Solution 8 — Timing error. Return →

The phase error is $1/120 = 0.0083$ cycles, or 3°. **Check:** a two-hour period is 120 minutes, so one minute is a small but measurable phase shift.

Solution 9 — Dominant term. Return →

The flat-field error at 0.05 mag dominates. Improve the flat-field calibration first. **Check:** reducing photon noise from 0.01 to 0.005 would barely change the total because the flat-field term is much larger.

Solution 10 — Discrepancy. Return →

The difference is 0.5. With uncertainty 0.2, this is 2.5σ, which is interesting and worth checking. With uncertainty 0.5, it is only 1σ and not significant. **Check:** significance depends on the uncertainty, not just the numerical difference.